

The K3 Stand: Neron–Severi Rank 20

forty-eight lines and the exact intersection matrix

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 8/16

1 Setting

The Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0 \subset \mathbb{P}^3$ is a K3 surface of maximal Néron–Severi rank. The Hodge conjecture for it is proved (Shioda, 1979); the program reproduces all invariants by exact integer arithmetic over $\mathbb{Q}$.

Theorem 1 (invariants of the K3 stand). *For the exact intersection matrix 49×49 (48 lines + the hyperplane class h): (1) $\text{rank}_{\mathbb{Q}} G = 20$ — matches Shioda’s $\rho = 20$; (2) the signature is $(1, 19)$; (3) the sublattice determinant is $64 = 8^2$; (4) the sum of the four lines of one family is equivalent to h .*

Proof. The lines: three families of 16: $L_1(a, b): x + i^a y = 0, z + i^b w = 0$; $L_2: x + i^a z = 0, y + i^b w = 0$; $L_3: x + i^a w = 0, y + i^b z = 0, a, b \in \mathbb{Z}/4$. The rules are combinatorial: $L \cdot L = -2$; within a family 1 iff exactly one index matches; between families cyclic conditions ($L_1 \cdot L_2: a_1 + b_2 \equiv a_2 + b_1$; $L_2 \cdot L_3: a_1 + b_1 \equiv a_2 + b_2$; $L_1 \cdot L_3: c \equiv a + b + d + 1$ — the extra unit from three odd exponents); $h \cdot L = 1$, $h^2 = 4$. Rank: exact fraction-free elimination — 20 nonzero rows; double registration via Smith minors. Signature: one positive, 19 negative eigenvalues, none zero (separation threshold 0.1). Determinant: the full K3 lattice is unimodular of signature $(3, 19)$; the sublattice index is 8, $\det = 64 = 8^2$. Equivalence: the intersection vector of the sum equals that of h componentwise (all 49); the difference is orthogonal to the algebraic sublattice and lies in $H^{2,0} \oplus H^{0,2}$ by the Hodge index theorem; but it is algebraic, and algebraic classes pair trivially with $H^{2,0}$; hence zero. $\square$

Proposition 2 (codimension-2 cycles). *The point class: $[\text{pt}] = h^2/4$ — exactly, since $(h^2)^2 = \deg X = 4$ and $H^4(X) = \mathbb{Q}h^2$. On $K3 \times K3$: the pairs (c_2, h_2) give 24 independent codim- $(2, 2)$ classes, the diagonal h^2 -pairs give 4 more; all algebraic with explicit supports. No blind spot: every $(2, 2)$ class orthogonal to them is zero by the Hodge index theorem.*

Protocol data: numbers

$n_{\text{lines}} = 48$; rank = 20 = ρ ; inertia $(1, 19)$; det = 64; hyperplane relation exact; 12 orbits; character multiplicities $(1, 7, 7, 7)$ summing to 22; codim 2: 4 + 24; runtime 1.04 s.

Summary of results: summary

All K3 invariants are computed exactly over $\mathbb{Q}$; the agreement with Shioda is confirmed; the family–hyperplane equivalence is proved exactly.

Verification: reproduction

`python3 laboratory.py` → “Stands → K3”: the matrix, rank, signature, determinant — within a second.

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